

Tutorial 2. Socket programming

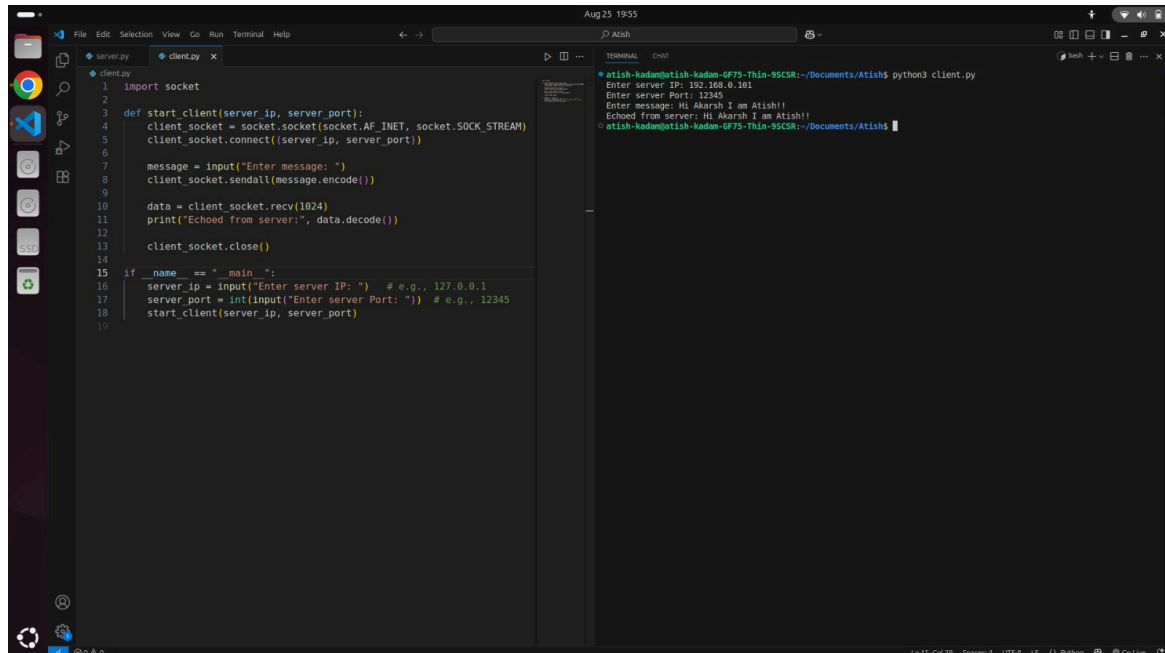
Name: Atish Kadam
Roll Number: CS25MTECH14003

Case 1.

Client - Atish Kadam(My Self)

Server - Akarsh Dubey (Friend)

Client Side- (IP - 192.168.0.103)

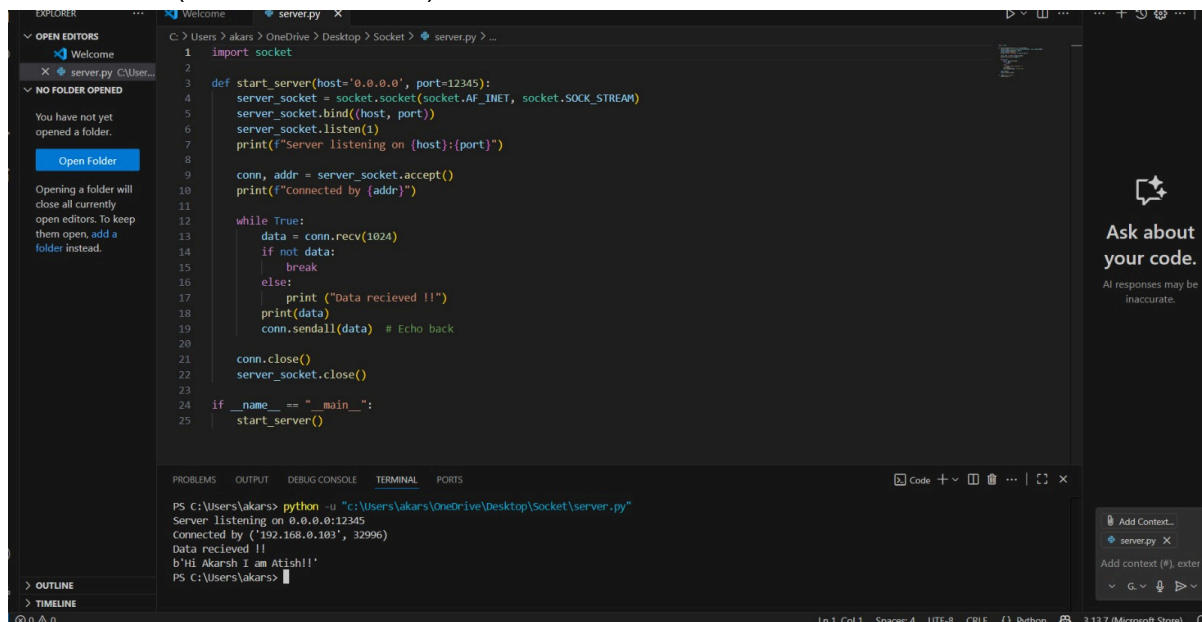


```
1 import socket
2
3 def start_client(server_ip, server_port):
4     client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
5     client_socket.connect((server_ip, server_port))
6
7     message = input("Enter message: ")
8     client_socket.sendall(message.encode())
9
10    data = client_socket.recv(1024)
11    print("Echoed from server:", data.decode())
12
13    client_socket.close()
14
15 if __name__ == "__main__":
16     server_ip = input("Enter server IP: ") # e.g., 127.0.0.1
17     server_port = int(input("Enter server Port: ")) # e.g., 12345
18     start_client(server_ip, server_port)
19
```

Terminal Output:

```
atish-kadam@atish-kadam-6f75-thin-95CSR:~/Documents/Atish$ python3 client.py
Enter server IP: 192.168.0.101
Enter server Port: 12345
Enter message: Hi Akarsh I am Atish!!
Echoed from server: Hi Akarsh I am Atish!!
atish-kadam@atish-kadam-6f75-thin-95CSR:~/Documents/Atish$
```

Server Side- (IP - 192.168.0.101)



```
1 import socket
2
3 def start_server(host='0.0.0.0', port=12345):
4     server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
5     server_socket.bind((host, port))
6     server_socket.listen(1)
7     print(f"Server listening on (host):(port)")
8
9     conn, addr = server_socket.accept()
10    print(f"connected by {addr}")
11
12    while True:
13        data = conn.recv(1024)
14        if not data:
15            break
16        else:
17            print("Data recieved !!")
18            print(data)
19            conn.sendall(data) # Echo back
20
21    conn.close()
22    server_socket.close()
23
24 if __name__ == "__main__":
25     start_server()
```

Terminal Output:

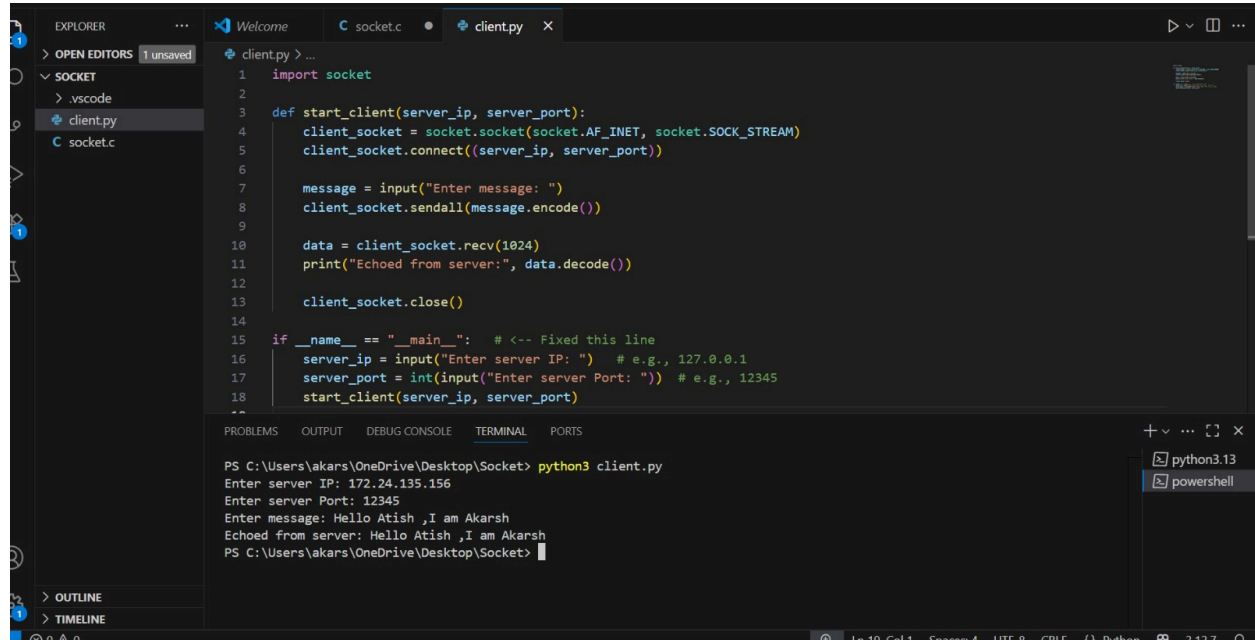
```
PS C:\Users\akars> python -u "c:\Users\akars\OneDrive\Desktop\Socket\server.py"
Server listening on 0.0.0.0:12345
connected by ('192.168.0.103', 32996)
Data recieved !!
b'Hi Akarsh I am Atish!!'
PS C:\Users\akars>
```

Case 2.

Server - Atish Kadam (My Self)

Client - Akarsh Dubey (Friend)

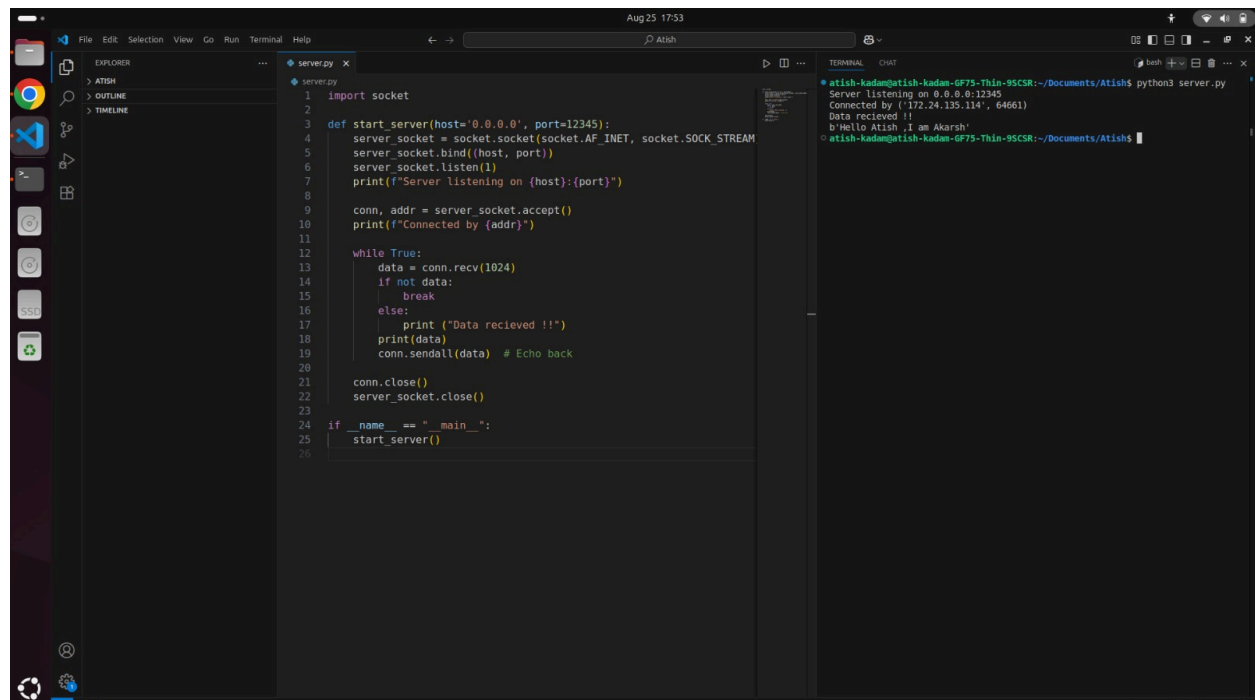
Client Side - (IP - 172.24.135.114)



The screenshot shows a VS Code editor with a file explorer on the left containing 'client.py' and 'socket.c'. The main editor displays the code for 'client.py', which imports the 'socket' module and defines a 'start_client' function. The function takes 'server_ip' and 'server_port' as arguments, creates a client socket, connects to the server, sends a message, receives a response, and prints it. The main block prompts the user for server IP and port, then for a message to send. The terminal at the bottom shows the execution of 'python3 client.py' with the following output:

```
PS C:\Users\akars\OneDrive\Desktop\Socket> python3 client.py
Enter server IP: 172.24.135.156
Enter server Port: 12345
Enter message: Hello Atish ,I am Akarsh
Echoed from server: Hello Atish ,I am Akarsh
PS C:\Users\akars\OneDrive\Desktop\Socket>
```

Server Side - (IP - 172.24.135.156)



The screenshot shows a VS Code editor with a file explorer on the left containing 'server.py'. The main editor displays the code for 'server.py', which imports the 'socket' module and defines a 'start_server' function. The function takes 'host' and 'port' as arguments, creates a server socket, binds it to the host and port, and starts listening for connections. It then enters a loop where it accepts connections, receives data, prints it, and sends it back to the client. The main block prompts the user for host and port, then starts the server. The terminal at the bottom shows the execution of 'python3 server.py' with the following output:

```
atish-kadam@atish-kadam-GF75-Thin-95CSR:~/Documents/Atish$ python3 server.py
Server listening on 0.0.0.0:12345
Connected by ('172.24.135.114', 64661)
Data recieved !!
b'Hello Atish ,I am Akarsh'
atish-kadam@atish-kadam-GF75-Thin-95CSR:~/Documents/Atish$
```